

Computer Vision Practical Report:

EDGE DETECTION USING SOBEL MATRIX ALONG Y AXIS

1. AIM

To detect horizontal edges in an image using the Sobel operator along the Y-axis in OpenCV, display the results, and save the output image.

2. PROCEDURE

- Load the image and convert it to grayscale.
- Apply Gaussian blur to reduce noise.
- Compute vertical gradients using `cv2.Sobel()` with `dx=0` and `dy=1`.
- Convert the gradient result to 8-bit format using `cv2.convertScaleAbs()`.
- Display the original and edge images using Matplotlib, and save the result using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2

image = cv2.imread(r"C:\Users\adminuser\Desktop\Python CV\E18-IMG.png")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

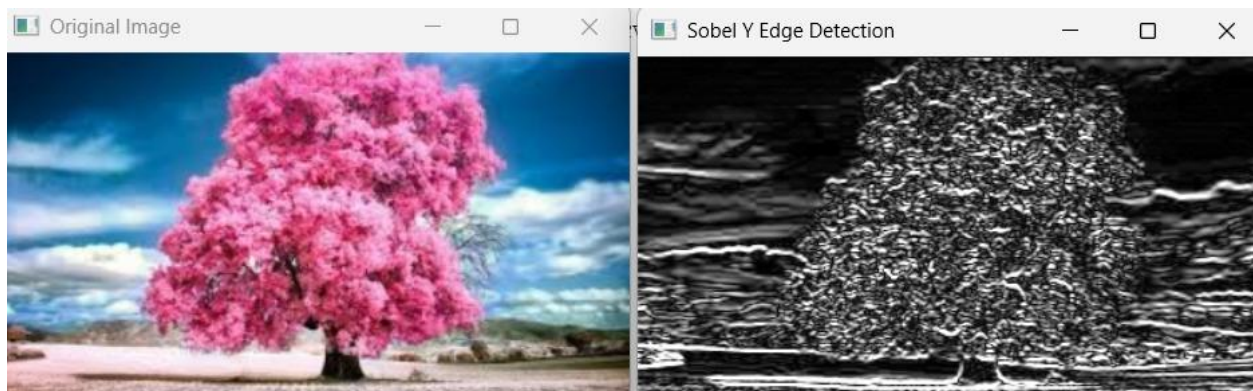
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_y = cv2.convertScaleAbs(sobel_y)

cv2.imshow("Original Image", image)
cv2.imshow("Sobel Y Edge Detection", sobel_y)

cv2.imwrite("sobel_y.jpg", sobel_y)
```

```
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.